🚕 TaxiPulse — The Bible Doc

Complete Project Reference & Interview Preparation Guide

📌 Table of Contents

1. Project Overview
2. Architecture Deep Dive
3. Data Flow — End to End
4. Medallion Architecture Explained
5. Tech Stack — What & Why
6. Database Schema & Star Schema
7. Data Quality Engine
8. Anomaly Detection System
9. Kafka Streaming vs Batch
10. Apache Airflow Orchestration
11. Docker Infrastructure
12. Streamlit Dashboard
13. Testing & CI/CD
14. How to Run the Project
15. Design Decisions & Tradeoffs
16. Production Readiness Discussion
17. How to Extend
18. Interview Q&A — 40+ Questions
19. 30-Second Pitch
20. Key Numbers to Remember
21. "How I Built This" — Journey & Problems Solved
22. Actual Performance Benchmarks
23. Code Walkthrough — Key Snippets Explained
24. Comparison — Why TaxiPulse Stands Out
25. Scenario-Based Interview Questions
26. Troubleshooting Guide
27. Project File Structure — Every File Explained
28. Environment Variables Reference
29. Skills Demonstrated — Component to Skill Map

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1. Project Overview

What is TaxiPulse?

TaxiPulse is an **end-to-end data engineering platform** that ingests, validates, transforms, and analyzes millions of real NYC taxi trip records. It demonstrates the complete lifecycle of a production data pipeline — from raw data ingestion through both batch and real-time streaming paths, to a star schema data warehouse with automated anomaly detection and interactive dashboards.

What Problem Does It Solve?

Imagine you're the data team at a ride-sharing company (Uber, Lyft). You need to:

- **Ingest millions of trip records** daily from multiple sources
- **Ensure data quality** — bad data costs millions in wrong pricing, incorrect analytics
- **Transform raw data** into clean, business-ready formats
- **Model data** so analysts can query it efficiently (not scanning millions of rows)
- **Detect anomalies** — if fares suddenly spike 300% in a zone, something's wrong
- **Alert the operations team** in real-time when anomalies are detected
- **Visualize everything** in dashboards for monitoring and decision-making

TaxiPulse does ALL of this, processing **9.5 million real NYC taxi trip records**.

The Data Source

NYC Taxi & Limousine Commission (TLC) Trip Record Data — publicly available from the NYC government. Contains actual taxi rides with:

- Pickup/dropoff datetime and location (zone IDs 1-263)
- Trip distance, fare amount, tip, tolls, total amount
- Payment type, rate code, passenger count
- Available as monthly Parquet files (each ~40-50 MB, ~3M rows)

We process **January, February, March 2024** — 3 months, 9,554,778 records total.

Key Metrics

Metric	Value
Total records processed	9,554,778
Records passed quality	9,417,374 (98.6%)
Records quarantined	137,403
Anomalies detected	3,623
Critical anomalies	1,478

Metric	Value
Data quality checks	18
Dimension tables	5
Aggregation tables	2
Hourly zone aggregations	240,716
Daily summaries	93
Docker services	8
Pytest tests	38
Airflow DAG tasks	7 (batch) + 2 (streaming)

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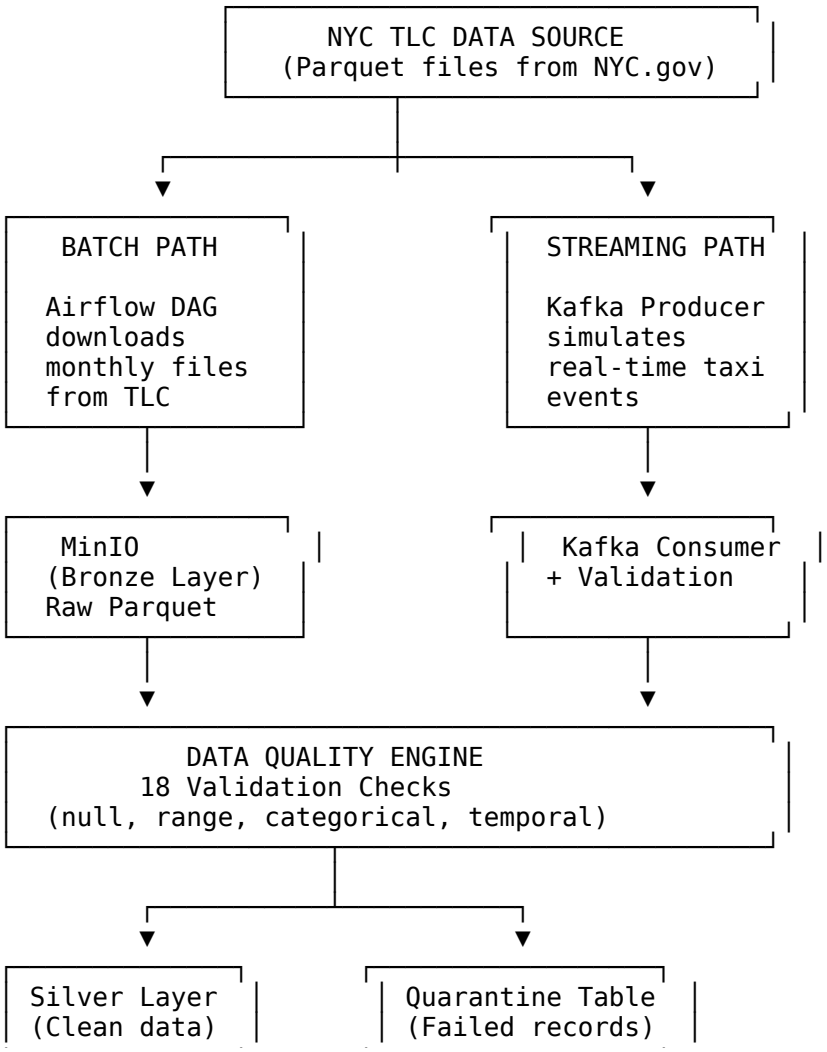
2. Architecture Deep Dive

High-Level Architecture



```
└──────────────┘        └──────────────────┘
        │
        ▼
┌──────────────────────────────────────────┐
│              GOLD LAYER                  │
│  Star Schema: fact_trips + 5 dimensions  │
│  Aggregations: hourly zone + daily summary │
└──────────────────────────────────────────┘
                    │
        ┌───────────┼───────────┐
        ▼           ▼           ▼
┌──────────────┐ ┌──────────────┐ ┌──────────────┐
│ Anomaly      │ │ Quality      │ │ Streamlit    │
│ Detection    │ │ Logging      │ │ Dashboard    │
│ + Slack      │ │              │ │ (4 pages)    │
└──────────────┘ └──────────────┘ └──────────────┘
```

Infrastructure Layer (All Dockerized)

```
Docker Compose (8 services)
├── PostgreSQL (Data Warehouse) — port 5432
├── PostgreSQL (Airflow Metadata) — port 5433
├── MinIO (S3-compatible Object Storage) — ports 9000, 9001
├── Zookeeper (Kafka dependency) — port 2181
├── Kafka (Streaming Platform) — ports 9092, 29092
├── Airflow Webserver (Pipeline UI) — port 8080
├── Airflow Scheduler (Executes DAGs)
└── Airflow Init (One-time DB setup)
```

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3. Data Flow — End to End

Batch Path (Primary)

```
Step 1: DOWNLOAD
  Airflow triggers Python script
  → Downloads Parquet files from https://d37ci6vzurychx.cloudfront.net/trip-
data/
  → Saves to local data/raw/ directory
  → Files: yellow_tripdata_2024-01.parquet, -02, -03

Step 2: UPLOAD TO MINIO (Bronze)
  Python script using minio library
  → Uploads Parquet files to MinIO bucket: taxipulse-data-lake/bronze/
  → This is the raw data lake — untouched source of truth

Step 3: LOAD TO POSTGRESQL (Bronze Table)
  Python script reads from MinIO
  → Reads Parquet with pandas/pyarrow
  → Standardizes column names
  → Inserts into bronze.raw_yellow_trips table
  → Batch inserts of 50,000 rows at a time
```

```
  → Adds metadata: load_timestamp, source_file

Step 4: DATA QUALITY VALIDATION
  Custom validation engine runs 18 checks:
  → 7 null checks (critical fields must not be null)
  → 6 range checks (fare $0-500, distance 0-200mi, etc.)
  → 3 categorical checks (payment_type 1-6, vendor 1-2)
  → 2 temporal checks (year 2020-2026, dropoff >= pickup)

  Results:
  → Clean records → Silver layer
  → Failed records → Quarantine table
  → Results logged to gold.quality_log (with JSON check details)

Step 5: SILVER TRANSFORMATION
  → Renames columns to standard names (PULocationID → pickup_location_id)
  → Enforces data types (ints, floats, timestamps)
  → Adds derived columns (trip_duration_minutes)
  → Removes exact duplicates
  → Loads into silver.clean_yellow_trips
  → Failed records → silver.quarantined_yellow_trips

Step 6: GOLD LAYER BUILD
  → Builds dim_datetime from unique pickup hours (2,185 rows)
  → Builds dim_pickup_location and dim_dropoff_location (262/261 zones)
  → dim_payment_type and dim_rate_code pre-loaded from init script
  → Builds fact_trips by joining Silver with dimensions (9,417,374 rows)
  → Builds agg_hourly_zone_revenue — pre-computed hourly stats per zone
(240,716 rows)
  → Builds agg_daily_summary — daily totals, averages, payment splits (93 rows)
  → All done via SQL INSERT...SELECT for performance (not Python row-by-row)

Step 7: ANOMALY DETECTION
  → Reads from Gold aggregation tables
  → Runs Z-score analysis on:
    - Fare anomalies (avg fare per zone/hour vs zone's historical mean)
    - Volume anomalies (trip count per zone/hour vs historical)
    - Daily revenue anomalies (total daily revenue vs historical)
  → Threshold: Z-score >= 3.0 (configurable)
  → Severity levels: critical (>=4), high (>=3.5), medium (>=3.0)
  → Logs to gold.anomaly_log
  → Sends Slack alerts for unsent anomalies
```

Streaming Path (Real-Time)

```
Step 1: KAFKA PRODUCER
  → Reads sample trips from Silver layer
  → Publishes each as a JSON event to Kafka topic "taxipulse-trips"
  → Adds metadata: event_id, event_timestamp, event_type
  → Uses pickup_location_id as partition key
  → In production: would be replaced by actual taxi dispatch events

Step 2: KAFKA CONSUMER
  → Subscribes to "taxipulse-trips" topic
  → Deserializes JSON events
  → Validates each event (required fields, range checks)
```

```
  → Valid events → INSERT into silver.clean_yellow_trips
  → Source marked as "kafka-stream" to distinguish from batch

Both paths feed into the SAME Silver table.
Gold layer and Anomaly Detection process all Silver data regardless of source.
```

Why Two Paths?

In real companies, data arrives in both ways:

- **Batch**: Historical data dumps, end-of-day reports, monthly files
- **Streaming**: Real-time events as they happen (ride completed, payment processed)

Having both demonstrates you understand the Lambda Architecture pattern — batch for completeness, streaming for real-time. The Silver layer unifies both.

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4. Medallion Architecture Explained

What is Medallion Architecture?

A data design pattern (popularized by Databricks) that organizes data into three layers based on quality and transformation level:

```
Bronze (Raw) → Silver (Cleaned) → Gold (Business-Ready)
```

Bronze Layer — "The Raw Vault"

Purpose: Store raw data exactly as received. No transformations. Source of truth.

In TaxiPulse:

- MinIO bucket: taxipulse-data-lake/bronze/ (Parquet files)
- PostgreSQL: bronze.raw_yellow_trips (9,554,778 rows)
- Columns match the original TLC schema exactly
- Includes metadata: load_timestamp, source_file

Why it matters: If a transformation bug corrupts Silver/Gold data, you can always re-process from Bronze. You never lose the original data.

Silver Layer — "The Clean Zone"

Purpose: Validated, cleaned, standardized data. One version of truth.

In TaxiPulse:

- silver.clean_yellow_trips (9,417,374 rows)
- silver.quarantined_yellow_trips (137,403 rows)

What happens here:

- Column names standardized (PULocationID → pickup_location_id)
- Data types enforced (strings → ints/floats/timestamps)
- Derived columns added (trip_duration_minutes)
- Duplicates removed
- Quality-failed records quarantined (not deleted — important!)

Who uses it: Data engineers for further processing, ML models for training.

Gold Layer — "The Business Layer"

Purpose: Business-ready data optimized for analytics and reporting.

In TaxiPulse:

- Star schema: gold.fact_trips + 5 dimension tables
- Aggregations: gold.agg_hourly_zone_revenue, gold.agg_daily_summary
- Logs: gold.anomaly_log, gold.quality_log

Key difference from Silver: Silver is a flat table with 9.4M rows. Gold is a dimensional model with pre-computed aggregations. Querying Gold is 100x faster for analytics.

Who uses it: Analysts, dashboards, reports, anomaly detection.

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5. Tech Stack — What & Why

Component	Tool	Why This Tool
Object Storage	MinIO	S3-compatible, runs locally in Docker, zero cost. Same API as AWS S3 — migrating to S3 is a config change
Data Warehouse	PostgreSQL	Industry-standard relational DB, perfect for star schema, excellent SQL support, free
Orchestration	Apache Airflow	Industry standard for batch pipeline orchestration. DAGs, scheduling, retries, monitoring UI
Streaming	Apache Kafka	Industry standard for event streaming. Decouples producers/consumers, handles backpressure
Containerization	Docker Compose	Entire infrastructure defined in one file. Anyone can clone and run with docker-compose up

Component	Tool	Why This Tool
Data Quality	Custom Python Engine	Built our own inspired by Great Expectations. No dependency conflicts, lightweight, configurable
Anomaly Detection	Python (scipy/numpy)	Z-score statistical analysis. Lightweight, no heavy ML framework needed
Alerting	Slack Webhooks	Free, instant notifications, easy integration
Dashboard	Streamlit	Python-native, deploys free to Streamlit Cloud, interactive charts with Plotly
Visualization	Plotly	Interactive charts, works seamlessly with Streamlit and pandas
Testing	pytest	Python standard for testing. Clean, simple, powerful
CI/CD	GitHub Actions	Free for public repos, runs tests on every push
Language	Python 3.11+	The lingua franca of data engineering

Why Local Docker Instead of Cloud?

Architecture matters more than the vendor.

MinIO is S3-compatible — the same boto3/minio code works with AWS S3 by changing one config line. PostgreSQL schemas map directly to BigQuery datasets or Redshift schemas. The engineering patterns are identical.

Benefits of local-first:

- Zero cost (no cloud bills)
- Anyone can clone and run the project
- Demonstrates deeper Docker/infrastructure understanding
- Cloud migration is a config change, not a rewrite

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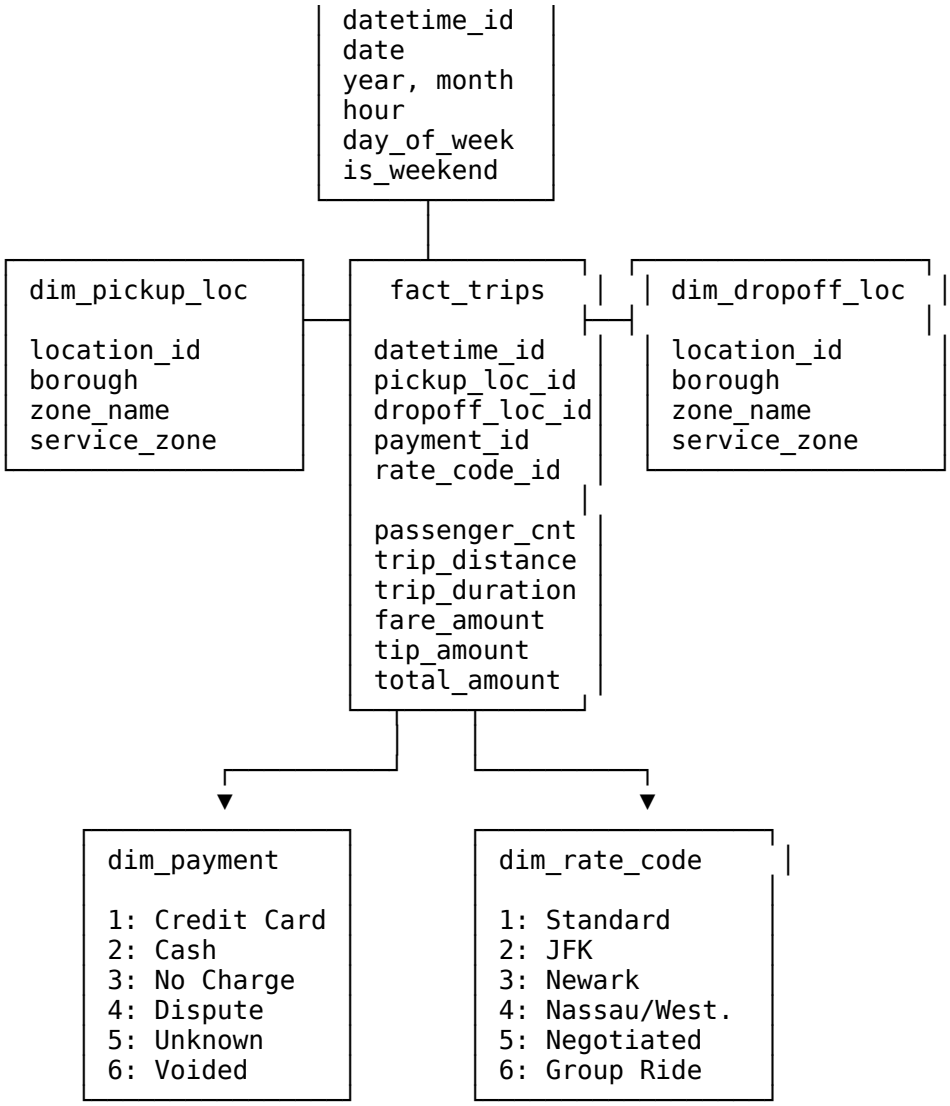
6. Database Schema & Star Schema

What is a Star Schema?

A dimensional modeling technique where a central **fact table** (containing measures/metrics) is surrounded by **dimension tables** (containing descriptive attributes). Named "star" because the diagram looks like a star.

```
┌──────────────┐
│ dim_datetime │
│              │
```



Why Star Schema vs Flat Table?

Aspect	Flat Table (Silver)	Star Schema (Gold)
Rows	9,417,374	9,417,374 (fact) + small dims
Query: "avg fare by hour"	Scans 9.4M rows every time (10+ sec)	Reads from agg table (milliseconds)
Payment type	1 (meaningless number)	"Credit Card" (via dimension join)
Location	zone_id = 132	"JFK Airport, Queens" (via dimension)
Storage	One massive table	Normalized, less redundancy
Use case	ML models, data engineers	Analysts, dashboards, reports

Complete Table List

Bronze Schema

Table	Rows	Purpose
raw_yellow_trips	9,554,778	Raw TLC data, untouched

Silver Schema

Table	Rows	Purpose
clean_yellow_trips	9,417,374	Cleaned, validated, enriched
quarantined_yellow_trips	137,403	Records that failed quality checks

Gold Schema

Table	Rows	Purpose
fact_trips	9,417,374	Central fact table with measures + dimension keys
dim_datetime	2,185	Date/time attributes (hour, day_of_week, is_weekend)
dim_pickup_location	262	Pickup zone names and boroughs
dim_dropoff_location	261	Dropoff zone names and boroughs
dim_payment_type	7	Payment method descriptions
dim_rate_code	8	Rate code descriptions
agg_hourly_zone_revenue	240,716	Pre-computed hourly stats per pickup zone
agg_daily_summary	93	Daily totals, averages, payment splits
anomaly_log	3,623	Detected anomalies with severity and z-scores
quality_log	3	Per-file validation results with JSON details

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7. Data Quality Engine

Why Data Quality Matters

Bad data → wrong analytics → wrong business decisions → lost money. In production:

- A negative fare could mean $millions in revenue miscalculation
- A trip distance of 9999 miles skews all averages
- A dropoff before pickup is physically impossible
- Missing location data means you can't analyze zone patterns

Our Validation Engine

Custom-built Python engine with **18 configurable expectations**:

Null Checks (7 — Critical)

Check	Column	Severity
pickup_datetime_not_null	tpep_pickup_datetime	Critical
dropoff_datetime_not_null	tpep_dropoff_datetime	Critical
fare_amount_not_null	fare_amount	Critical
total_amount_not_null	total_amount	Critical
trip_distance_not_null	trip_distance	Critical
pu_location_not_null	pu_location_id	Critical
do_location_not_null	do_location_id	Critical

Range Checks (6 — Critical)

Check	Column	Range	Severity
fare_amount_range	fare_amount	$0 — $500	Critical
total_amount_range	total_amount	$0 — $1000	Critical
trip_distance_range	trip_distance	0 — 200 miles	Critical
passenger_count_range	passenger_count	0 — 9	Warning
tip_amount_range	tip_amount	$0 — $200	Critical
tolls_amount_range	tolls_amount	$0 — $200	Critical

Categorical Checks (3 — Warning)

Check	Column	Valid Values	Severity
payment_type_valid	payment_type	1,2,3,4,5,6	Warning
vendor_id_valid	vendor_id	1,2	Warning
ratecode_valid	ratecode_id	1,2,3,4,5,6,99	Warning

Temporal Checks (2 — Critical)

Check	Description	Severity
pickup_year_valid	Year between 2020-2026	Critical
dropoff_after_pickup	Dropoff time >= pickup time	Critical

Severity System

- **Critical**: Record is quarantined — NOT promoted to Silver
- **Warning**: Record is flagged but still promoted — logged for review

Results on Real Data

Metric	Value
Total records	9,554,778
Passed all critical checks	9,417,375 (98.6%)
Quarantined	137,403 (1.4%)
Top quarantine reason	Fare/total out of range (107,665)
Second reason	Tolls out of range (7,564)

Quality Logging

Every validation run is logged to gold.quality_log with:

- Source file name
- Total/passed/failed counts
- Pass rate percentage
- Full JSON details of every check result

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8. Anomaly Detection System

How It Works

Uses **Z-score analysis** on the Gold aggregation tables to detect statistically unusual patterns.

What is Z-score?

Z-score measures how many standard deviations a value is from the mean:

```
Z = (value - mean) / standard_deviation
```

- Z-score of 0 = exactly average
- Z-score of 1 = one standard deviation above mean
- Z-score of 3 = three standard deviations above mean (very unusual, ~0.3% probability)

Three Types of Anomaly Detection

1. Fare Anomalies (2,106 detected)

- Analyzes: average fare per pickup zone per hour
- Groups by: pickup_location_id
- Detects: zones where the average fare is abnormally high or low for that zone's historical pattern
- Example: Zone 48 avg fare $81.36 at 4AM (normally ~$15) → Z-score 20.6

2. Volume Anomalies (1,515 detected)

- Analyzes: trip count per pickup zone per hour
- Groups by: pickup_location_id
- Detects: unusual spikes or drops in trip volume
- Example: Zone 226 had 53 trips in one hour (normally ~5) → Z-score 16.1

3. Daily Revenue Anomalies (2 detected)

- Analyzes: total daily revenue across all zones
- No grouping — looks at overall daily patterns
- Detects: days where total revenue is significantly different from the norm

Severity Classification

Severity	Z-Score Threshold	Count
🔴 Critical	>= 4.0	1,478
🟠 High	>= 3.5	781
🟡 Medium	>= 3.0	1,364

Alerting

When anomalies are detected:

1. All anomalies are logged to gold.anomaly_log table
2. Unsent anomalies are sent to Slack via webhook
3. Formatted message with severity, description, zone info
4. Marked as alert_sent = TRUE after successful delivery
5. Falls back to console logging if Slack is not configured

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9. Kafka Streaming vs Batch

Why Both?

In real companies, data arrives in two ways:

Aspect	Batch	Streaming
Frequency	Periodic (hourly/daily/monthly)	Continuous (real-time)
Data source	File dumps, database exports	Event streams, APIs
Latency	Minutes to hours	Seconds to milliseconds
Volume per event	Millions of rows at once	One event at a time
Use case	Historical analysis, backfills	Real-time monitoring, alerts

How They Work Together (Lambda Architecture)

```
  Batch (monthly):  TLC Files ──────────→┌──────────────┐
                                         │   Silver     │──→ Gold ──→ Anomaly
Detection
  Streaming (24/7): Kafka events ───────→│   Layer      │        (runs on batch
schedule)
                                         └──────────────┘
```

Both paths feed into the **same Silver table**. The Gold layer doesn't care where the data came from — it processes all Silver data. This is the beauty of the Medallion architecture.

Batch Path in TaxiPulse

- **Orchestration:** Apache Airflow (scheduled, 7-task DAG)
- **Ingestion:** Downloads monthly Parquet files → MinIO → PostgreSQL
- **Processing:** Validation → Silver → Gold → Anomaly Detection
- **Frequency:** Monthly (but can be configured to any schedule)

Streaming Path in TaxiPulse

- **Producer:** Reads sample trips, publishes to Kafka topic as JSON events
- **Consumer:** Subscribes to topic, validates each event, inserts to Silver
- **Partition key:** pickup_location_id (ensures zone ordering)
- **In production:** Producer would be replaced by actual taxi dispatch system

Why Two Airflow DAGs?

DAG	Schedule	Tasks	Purpose
taxipulse_batch_pipeline	@monthly	7	Full ETL: download → upload → bronze → validate → silver → gold → anomaly
taxipulse_streaming_demo	Manual trigger	2	Demo: produce Kafka events → consume & insert to Silver

The streaming DAG is for **demonstration**. In production, Kafka consumers run as **always-on services** (e.g., Kubernetes pods), not as Airflow tasks. Airflow is designed for batch orchestration, not long-running streaming processes.

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10. Apache Airflow Orchestration

Batch Pipeline DAG

```
download_tlc_data → upload_to_minio_bronze → load_bronze_to_postgres
    → validate_data_quality → transform_to_silver → build_gold_layer
    → run_anomaly_detection
```

Task Details

Task	What It Does	Failure Handling
download_tlc_data	Downloads Parquet files from NYC TLC	Retries 2x, skips already-downloaded files
upload_to_minio_bronze	Uploads to MinIO data lake	Retries 2x, skips already-uploaded files
load_bronze_to_postgres	Reads from MinIO, inserts to PostgreSQL Bronze	Skips already-loaded files (checks source_file)
validate_data_quality	Runs 18 quality checks	Fails if pass rate < 80%
transform_to_silver	Cleans, enriches, loads Silver + quarantine	Skips if Silver already has data
build_gold_layer	Builds star schema + aggregations via SQL	Skips already-populated tables
run_anomaly_detection	Z-score analysis + Slack alerts	Logs anomalies regardless of alert success

DAG Configuration

- **Schedule:**@monthly
- **Retries:** 2 per task
- **Retry delay:** 5 minutes
- **Catchup:** False (don't backfill missed runs)
- **Executor:** LocalExecutor

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11. Docker Infrastructure

Service Architecture

```
8 Docker Services:
  postgres:          Data warehouse (port 5432)
  postgres-airflow:  Airflow metadata DB (port 5433)
  minio:             Object storage (ports 9000, 9001)
  zookeeper:         Kafka coordination (port 2181)
  kafka:             Streaming platform (ports 9092, 29092)
  airflow-webserver: Pipeline UI (port 8080)
  airflow-scheduler: Executes DAG tasks
  airflow-init:      One-time DB setup + admin user creation
```

Why Separate PostgreSQL Instances?

Best practice — Airflow's internal metadata (task states, DAG runs, connections) should NOT share a database with your application data. Keeps concerns separated and prevents Airflow operations from affecting your warehouse.

Volume Persistence

All data is stored in named Docker volumes:

- taxipulse-postgres-data — warehouse data survives restarts
- taxipulse-minio-data — data lake files persist
- taxipulse-kafka-data — Kafka messages preserved

docker-compose down preserves data. docker-compose down -v destroys everything (clean start).

Health Checks

PostgreSQL and MinIO have health checks configured:

- PostgreSQL: pg_isready command
- MinIO: HTTP health endpoint check
- Other services depend on these health checks before starting

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12. Streamlit Dashboard

Architecture

Two deployment modes:

1. **Local:** Connects directly to PostgreSQL (full SQL queries)
2. **Cloud:** Reads from pre-exported CSV files (zero database needed)

db.py has named functions (get_bronze_count(), get_daily_summary(), etc.) that automatically route to the correct data source.

Four Pages

📊 Pipeline Overview

- Key metrics: Bronze/Silver/Quarantine/Anomaly counts
- Data quality pass rate progress bar
- Gold layer table counts
- Pipeline architecture diagram
- Data freshness (earliest/latest dates, days covered)

🗺️ Analytics Explorer

- Daily revenue trend (area chart)
- Daily trip volume (bar chart)
- Hourly trip patterns (bar + line charts)
- Top 20 pickup zones with real names (bar + scatter)
- Payment method distribution over time
- Summary statistics (avg trips, revenue, fare, distance)

🚨 Anomaly Monitor

- Summary metrics (total, critical, high, medium, alerts sent)
- Interactive filters (severity, anomaly type)
- Distribution charts (by type, by severity)
- Top 15 anomalous zones with names
- Z-score distribution histogram
- Recent anomalies data table

✅ Data Quality Report

- Overall metrics (total, passed, failed, pass rate)
- Quality by source file (stacked bar + pass rate bar)
- Pass/fail pie chart
- Per-check results (bar chart + detailed table)
- Quarantine summary (count, files affected)
- Top quarantine reasons (horizontal bar)

Live URL

https://taxipulse-srujankothuri.streamlit.app

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13. Testing & CI/CD

Test Coverage

38 tests across 4 modules:

test_ingestion (5 tests)

- URL format validation
- Filename generation
- Default config values
- Month differentiation

test_quality (13 tests)

- Null checks (all valid, with nulls, all null)
- Range checks (in range, below, above, null handling)
- Categorical checks (valid, invalid, null)
- Temporal checks (valid year, invalid year, dropoff order)
- Full validation (valid data, negative fare quarantine, expectations count, empty df)

test_anomaly (6 tests)

- Normal data detection
- Obvious outlier detection
- Grouped detection
- Empty dataframe handling
- Single value handling
- Threshold sensitivity

test_transformations (8 tests)

- Column renaming
- Unmapped column preservation
- Integer type enforcement
- Datetime parsing
- Trip duration calculation
- Duration capping (max 24h)
- Negative duration capping
- Duplicate removal

GitHub Actions CI/CD

Two jobs run on every push to main:

1. Test Job:

- Sets up Python 3.11
- Installs dependencies
- Creates .env from template
- Runs all 38 pytest tests

2. Lint Job:

- Compiles key Python files to check for syntax errors
- Validates settings, constants, quality, anomaly modules

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14. How to Run the Project

Prerequisites

- Docker Desktop (4GB+ RAM allocated in Settings → Resources)
- Python 3.11+ (check: python --version)
- Git (check: git --version)
- pip (check: pip --version)

Step-by-Step Setup (From Scratch)

1. Clone the Repository

```
git clone https://github.com/srujankothuri/TaxiPulse.git
cd TaxiPulse
```

2. Create Python Virtual Environment

```
python -m venv venv

# Activate (Mac/Linux):
source venv/bin/activate

# Activate (Windows):
venv\Scripts\activate

# Verify:
which python   # Should show .../TaxiPulse/venv/bin/python
```

3. Install Python Dependencies

```
pip install --upgrade pip
pip install -r requirements.txt
pip install psycopg2-binary pytest streamlit plotly sqlalchemy kafka-python-ng
        minio loguru
```

4. Configure Environment

```
cp .env.example .env
```

Then edit .env — change these three lines for local use:

```
MINIO_ENDPOINT=localhost:9000
POSTGRES_HOST=localhost
KAFKA_BOOTSTRAP_SERVERS=localhost:29092
```

Everything else stays as default.

5. Start Docker Infrastructure

```
# Make sure Docker Desktop is running first!
docker-compose up -d
```

First time takes 5-10 minutes (downloads images). Check status:

```
docker-compose ps
```

All 7+ services should show Up. Wait ~60 seconds for Airflow to initialize.

6. Verify Services Are Running

Service	How to Verify
PostgreSQL	docker exec -it taxipulse-postgres psql -U taxipulse -d taxipulse -c "" → Should show bronze, silver, gold schemas
MinIO	Open http://localhost:9001 → Login: taxipulse / taxipulse123
Airflow	Open http://localhost:8080 → Login: admin / admin
Kafka	`docker-compose ps

7. Run the Complete Pipeline

Option A: One command (using Makefile)

```
make pipeline
```

This runs: ingestion → quality → silver → gold → anomaly (takes ~50 minutes total)

Option B: Step by step (recommended first time)

```
# Step 1: Download NYC data + upload to MinIO + load to Bronze (~15 min)
python scripts/run_ingestion.py

# Step 2: Run data quality validation (~3 min)
python scripts/run_quality_check.py

# Step 3: Transform to Silver layer (~25 min — longest step)
python scripts/run_silver.py

# Step 4: Build Gold star schema + aggregations (~5 min)
python scripts/run_gold.py

# Step 5: Run anomaly detection + send alerts (~30 sec)
python scripts/run_anomaly_detection.py

# Step 6: Load NYC zone names for readable dashboards (~10 sec)
python scripts/load_zone_names.py
```

8. Run Kafka Streaming Demo

```
python scripts/run_streaming_demo.py
```

Produces 500 events to Kafka, consumes them, validates, and inserts to Silver. Takes ~30 seconds.

9. Launch Dashboard

```
python -m streamlit run streamlit_app/app.py
```

Opens at http://localhost:8501 — Navigate through all 4 pages.

10. Run Tests

```
python -m pytest tests/ -v
```

Should show 38 passed.

All Available Commands (Makefile)

Run make help to see all commands:

```
make help              # Show all available commands

# --- Infrastructure ---
make up                # Start all Docker services
make down              # Stop all Docker services
```

```
make restart           # Restart all Docker services
make ps                # Show running containers
make logs s=kafka      # Show logs for a specific service

# --- Pipeline Steps (Individual) ---
make ingestion         # Download + upload to MinIO + load Bronze
make quality           # Run data quality validation
make silver            # Run Silver layer transformation
make gold              # Build Gold star schema + aggregations
make anomaly           # Run anomaly detection + alerting
make streaming         # Run Kafka streaming demo (500 events)
make zones             # Load NYC zone names into dimension tables

# --- Full Pipeline ---
make pipeline          # Run complete batch pipeline (ingestion → anomaly)

# --- Dashboard ---
make dashboard         # Launch Streamlit dashboard (http://localhost:8501)

# --- Testing ---
make test              # Run all 38 pytest tests

# --- Data Export ---
make export-data       # Export Gold tables to CSV for Streamlit Cloud

# --- Cleanup ---
make clean             # Stop services + delete all data volumes (DESTRUCTIVE!)
make clean-python      # Remove Python cache files (__pycache__)
```

All Runner Scripts (in scripts/ folder)

Script	What It Does	Run Time
scripts/run_ingestion.py	Downloads 3 months of NYC data, uploads to MinIO, loads to Bronze PostgreSQL	~15 min
scripts/run_quality_check.py	Runs 18 validation checks on 9.5M Bronze rows, logs results	~3 min
scripts/run_silver.py	Validates + transforms + loads 9.4M rows to Silver, quarantines 137K	~25 min
scripts/run_gold.py	Builds star schema (fact + dimensions) + aggregation tables via SQL	~5 min
scripts/run_anomaly_detection.py	Z-score anomaly detection on Gold tables + Slack alerts	~30 sec
scripts/run_streaming_demo.py	Kafka produce 500 events → consume → validate → insert to Silver	~30 sec
scripts/load_zone_names.py	Downloads NYC zone lookup CSV, updates dimension tables with names	~10 sec

---

All Access Points

Service	URL	Credentials	What You See
Streamlit Dashboard	http://localhost:8501	None	4-page analytics dashboard
Airflow UI	http://localhost:8080	admin / admin	DAGs, task runs, logs
MinIO Console	http://localhost:9001	taxipulse / taxipulse123	Data lake buckets (bronze/)
PostgreSQL	localhost:5432	taxipulse / taxipulse123	Data warehouse (bronze/silver/gold schemas)
Live Dashboard	https://taxipulse-srujankothuri.streamlit.app	None	Deployed version (CSV mode)

Useful PostgreSQL Commands

```
# Connect to PostgreSQL
docker exec -it taxipulse-postgres psql -U taxipulse -d taxipulse

# Inside psql:
\dn                     # List schemas (bronze, silver, gold)
\dt bronze.*            # List Bronze tables
\dt silver.*            # List Silver tables
\dt gold.*              # List Gold tables
\d gold.fact_trips      # Describe a table's columns

# Quick counts:
SELECT 'bronze' as layer, COUNT(*) FROM bronze.raw_yellow_trips
UNION ALL SELECT 'silver', COUNT(*) FROM silver.clean_yellow_trips
UNION ALL SELECT 'quarantine', COUNT(*) FROM silver.quarantined_yellow_trips
UNION ALL SELECT 'fact_trips', COUNT(*) FROM gold.fact_trips
UNION ALL SELECT 'anomalies', COUNT(*) FROM gold.anomaly_log;

# Sample data:
SELECT * FROM silver.clean_yellow_trips LIMIT 5;
SELECT * FROM gold.agg_daily_summary ORDER BY date LIMIT 10;
SELECT * FROM gold.anomaly_log ORDER BY z_score DESC LIMIT 10;

# Zone names:
SELECT pickup_location_id, zone_name, borough
FROM gold.dim_pickup_location
WHERE zone_name IS NOT NULL
ORDER BY pickup_location_id LIMIT 20;

# Quality results:
SELECT source_file, total_records, passed_records, failed_records, pass_rate
```

```
FROM gold.quality_log;

# Exit psql:
\q
```

Common Workflows

"I want to start from scratch"

```
docker-compose down -v          # Stop everything, delete all data
docker-compose up -d            # Start fresh
# Wait 60 seconds...
make pipeline                   # Run full pipeline
python scripts/load_zone_names.py
make dashboard                  # Launch dashboard
```

"I want to add more months of data"

Edit .env:

```
TLC_DATA_MONTHS=01,02,03,04,05,06
```

Then:

```
# Need to truncate Silver/Gold first (they have skip-if-loaded logic)
docker exec -it taxipulse-postgres psql -U taxipulse -d taxipulse -c "
  TRUNCATE silver.clean_yellow_trips CASCADE;
  TRUNCATE silver.quarantined_yellow_trips;
  TRUNCATE gold.fact_trips CASCADE;
  TRUNCATE gold.dim_datetime CASCADE;
  TRUNCATE gold.agg_hourly_zone_revenue;
  TRUNCATE gold.agg_daily_summary;
  TRUNCATE gold.anomaly_log;
  TRUNCATE gold.quality_log;
  TRUNCATE bronze.raw_yellow_trips;
"
make pipeline
python scripts/load_zone_names.py
```

"I want to re-export CSVs for Streamlit Cloud"

```
make export-data
git add -f streamlit_app/data/*.csv
git commit -m "Update exported CSV data"
git push
# Streamlit Cloud auto-redeploys in 2-3 minutes
```

"Pipeline crashed midway — how do I resume?"

Just re-run the same command. Every step has idempotency checks:

- Download: skips already-downloaded files
- MinIO upload: skips already-uploaded objects
- Bronze load: skips already-loaded source files
- Silver: skips if table already has data
- Gold: skips if tables already populated

"Docker is eating too much disk space"

```
docker system df                    # Check usage
docker system prune -a              # Remove unused images/containers
docker volume prune                 # Remove unused volumes (careful!)
```

"Tests are failing after code changes"

```
python -m pytest tests/ -v --tb=long    # Verbose output with full tracebacks
python -m pytest tests/test_quality/ -v  # Run just one test module
python -m pytest tests/ -k "test_null"   # Run tests matching a keyword
```

---

15. Design Decisions & Tradeoffs

Decision 1: Local Docker vs Cloud

Chose: Local Docker with MinIO + PostgreSQL**Over:** GCP (BigQuery + GCS)**Reason:** GCP billing verification issues. Local-first approach means anyone can run the project. Architecture is cloud-agnostic — MinIO is S3-compatible, PostgreSQL schemas map to any warehouse.**Tradeoff:** No Terraform IaC (Docker Compose serves the same purpose locally).

Decision 2: Custom Quality Engine vs Great Expectations

Chose: Custom Python validation engine**Over:** Great Expectations library**Reason:** GE had severe dependency conflicts with pandas/numpy versions. Custom engine is lighter, zero conflicts, fully configurable. Same validation patterns (expectations, severity levels, quarantining).**Tradeoff:** Less feature-rich than full GE, but demonstrates deeper understanding.

Decision 3: SQL-based Gold Transformations vs Python/dbt

Chose: SQL INSERT...SELECT statements**Over:** dbt or Python row-by-row processing**Reason:** Direct SQL is fastest for PostgreSQL (no data movement through Python). Building fact_trips via JOIN is a single SQL statement that processes 9.4M rows in minutes.**Tradeoff:** Less modular than dbt, but simpler for a portfolio project.

Decision 4: Z-score vs ML-based Anomaly Detection

Chose: Z-score statistical method**Over:** Isolation Forest, LSTM, or other ML approaches**Reason:** Z-score is interpretable, fast, and appropriate for this data. ML-based approaches need training data and are overkill for detecting fare/volume spikes. Z-score clearly explains "this value is X standard deviations from normal."**Tradeoff:** Less sophisticated than ML, but more explainable and faster.

Decision 5: Simulated Streaming vs Real-time API

Chose: Kafka producer that simulates events from historical data**Over:** Connecting to a real-time data source**Reason:** NYC TLC data is released monthly, not in real-time. Simulation demonstrates the exact same engineering pattern (producer → Kafka → consumer → database) that would work with a real event stream.**Tradeoff:** Not "truly real-time" but architecturally identical.

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16. Production Readiness Discussion

What's Production-Ready

✅ Containerized infrastructure (Docker Compose → Kubernetes) ✅ Idempotent pipeline (re-running doesn't duplicate data) ✅ Data quality checks with quarantine ✅ Error handling and retry logic ✅ Logging throughout ✅ Health checks on services ✅ CI/CD pipeline ✅ Tests ✅ Alerting (Slack)

What Would Change for Production

Aspect	Current (Demo)	Production
Orchestration	Airflow on Docker Compose	Airflow on Kubernetes (MWAA/Cloud Composer)
Storage	MinIO	AWS S3 or GCS
Warehouse	PostgreSQL	Snowflake, BigQuery, or Redshift
Streaming	Kafka on Docker	Confluent Cloud or Amazon MSK
Kafka Consumer	Airflow task (demo)	Kubernetes pod (always-on)

Aspect	Current (Demo)	Production
Monitoring	Streamlit + Slack	Datadog/Grafana + PagerDuty
Scale	Single node	Distributed (Spark for transforms)
Auth	Default passwords	Secrets Manager + IAM
Data	3 months	Years of data, incremental loads

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17. How to Extend

1. **Add more data months** — Change TLC_DATA_MONTHS in .env
2. **Add dbt** — Replace SQL transformations with dbt models for versioning/testing
3. **Add Spark** — For processing larger datasets that don't fit in memory
4. **Add Grafana** — For infrastructure monitoring alongside Streamlit
5. **Add CDC (Change Data Capture)** — Use Debezium to capture database changes
6. **Add data lineage** — Track how data flows from source to dashboard
7. **Cloud deployment** — Terraform scripts for AWS/GCP infrastructure
8. **ML anomaly detection** — Add Isolation Forest for comparison with Z-score

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18. Interview Q&A — 40+ Questions

Architecture & Design

Q1: Walk me through your architecture.

> "TaxiPulse is an end-to-end data engineering platform with dual ingestion paths. The batch path uses Airflow to orchestrate downloading NYC taxi data, uploading to MinIO as a data lake, and loading into PostgreSQL. The streaming path uses Kafka to simulate real-time taxi events. Both paths feed into a unified Silver layer after data quality validation. The Gold layer contains a star schema with pre-computed aggregations, and an anomaly detection engine monitors for fare and volume spikes, sending Slack alerts. Everything is containerized with Docker Compose."

Q2: Why did you choose Medallion Architecture?

> "Medallion Architecture gives me clear separation of concerns. Bronze preserves raw data as a source of truth — if I ever need to reprocess, I start from Bronze. Silver is the cleaned, validated version that data engineers and ML models use. Gold is optimized for analysts and dashboards with a star schema and pre-aggregated tables. This pattern is used by Databricks, Netflix, and most modern data platforms."

Q3: Why MinIO instead of S3?

> "MinIO is S3-compatible — it implements the exact same API. My minio/boto3 code would work with AWS S3 by changing one endpoint config. I chose local-first to avoid cloud costs and make the project runnable by anyone with just Docker. The architecture is cloud-agnostic by design."

Q4: Why PostgreSQL instead of BigQuery/Snowflake?

> "Same reasoning — PostgreSQL demonstrates the same dimensional modeling, SQL optimization, and schema design skills. The star schema I built would translate directly to BigQuery or Snowflake. PostgreSQL runs in Docker with zero cost, and anyone can clone the repo and run the project."

Q5: How would you deploy this on AWS?

> "I'd replace MinIO with S3, PostgreSQL with Redshift Serverless or RDS, Kafka with Amazon MSK or Confluent Cloud, and Airflow with Amazon MWAA. Terraform scripts would provision all resources. The Python code stays almost identical — just changing connection configs."

Data Quality

Q6: How does your data quality engine work?

> "I built a custom validation engine with 18 configurable expectations across four categories: null checks, range checks, categorical checks, and temporal checks. Each expectation has a severity — critical or warning. Critical failures quarantine the record, warnings are logged but the record passes. Results are stored in a quality_log table with JSON details for each check. The engine processed 9.5 million rows and achieved a 98.6% pass rate."

Q7: What happens when data quality fails?

> "Failed records are NOT deleted — they're quarantined to a separate table (silver.quarantined_yellow_trips) with the failure reason attached. This is important because you might want to investigate why records failed, fix upstream issues, or adjust thresholds. Deletion would lose that information forever."

Q8: Why did you build a custom engine instead of using Great Expectations?

> "I initially planned to use Great Expectations, but it had severe dependency conflicts with the pandas and numpy versions required by other components. Rather than fighting dependency hell, I built a lightweight custom engine that implements the same patterns — configurable expectations, severity levels, quarantining, and result logging. This actually demonstrates deeper understanding of data quality principles than just importing a library."

Q9: What's your 98.6% pass rate mean?

> "Out of 9,554,778 raw records, 9,417,375 passed all critical quality checks and were promoted to Silver. 137,403 were quarantined — mostly due to negative fares or totals exceeding $1000, which are real data quality issues in the NYC TLC dataset. The 98.6% is per-record across all critical checks combined."

Anomaly Detection

Q10: How does your anomaly detection work?

> "I use Z-score analysis on the Gold aggregation tables. For each pickup zone, I calculate the mean and standard deviation of average fares across all hours. When a specific hour's average fare is more than 3 standard deviations from that zone's mean, it's flagged as an anomaly. I do the same for trip volume and daily revenue. The threshold is configurable — 3.0 by default, with severity levels at 3.0 (medium), 3.5 (high), and 4.0 (critical)."

Q11: Why Z-score instead of machine learning?

> "Z-score is interpretable, fast, and appropriate for this use case. When I flag 'Zone 48 avg fare $81 at 4AM, z-score 20.6,' anyone understands that $81 is 20.6 standard deviations above normal — extremely unusual. ML methods like Isolation Forest are more powerful for complex patterns, but for detecting fare and volume spikes, statistical methods are more explainable and don't require training data."

Q12: What kinds of anomalies did you find?

> "3,623 total — 2,106 fare spikes, 1,515 volume spikes, and 2 daily revenue anomalies. The most severe was Zone 48 with an average fare of $81.36 at 4AM (z-score 20.6). This makes sense — late-night trips from certain zones like airports or outer boroughs tend to have much higher fares. The system correctly identifies these as statistically unusual for those zones."

Streaming

Q13: How does your streaming path differ from batch?

> "Batch processes data in bulk — downloading entire monthly files and processing millions of rows at once through the full pipeline. Streaming processes individual events in real-time through Kafka. The key difference is latency: batch has minutes-to-hours latency, streaming has seconds latency. Both paths feed into the same Silver table, which is the unifying layer."

Q14: Why isn't the Kafka consumer in the batch Airflow DAG?

> "Airflow is designed for batch orchestration — scheduled tasks that run and finish. Kafka consumers are long-running processes that need to be always-on. In production, the consumer would run as a Kubernetes pod, not an Airflow task. I have a separate streaming demo DAG for testing purposes, but that's for demonstration only."

Q15: Is your streaming really real-time?

> "The Kafka architecture is genuinely real-time — producer publishes events, consumer processes them within seconds. The data itself is simulated from historical records because NYC TLC doesn't provide a real-time feed. But the engineering pattern is identical to what Uber or Lyft uses — replace my producer with their dispatch event system and the rest works unchanged."

Star Schema & Modeling

Q16: Explain your star schema design.

> "The central fact table (fact_trips) contains measures like fare_amount, trip_distance, and tip_amount, plus foreign keys to five dimension tables. dim_datetime captures time attributes (hour, day_of_week, is_weekend), dim_pickup_location and dim_dropoff_location map zone IDs to borough names and zone names, dim_payment_type translates codes to descriptions, and dim_rate_code describes fare calculation methods. I also have two pre-aggregated tables for dashboard performance."

Q17: Why pre-computed aggregations?

> "Without aggregations, every dashboard query would scan 9.4 million rows — slow and resource-intensive. agg_hourly_zone_revenue (240K rows) and agg_daily_summary (93 rows) contain pre-computed totals, averages, and counts. Dashboard queries hit these small tables and return in milliseconds instead of seconds."

Q18: Why separate pickup and dropoff dimension tables?

> "Because a trip has two different location contexts — where it started and where it ended. While the underlying zone data is the same, having separate dimension tables allows independent analysis. For example, 'zones with the most pickups' is different from 'zones with the most dropoffs.' It also follows star schema best practices of one dimension per analytical context."

Infrastructure

Q19: Why Docker Compose instead of Kubernetes?

> "Docker Compose is the right tool for a portfolio project — it's simple, portable, and anyone can run it with a single command. Kubernetes would add significant complexity (manifests, helm charts, ingress controllers) without adding value for demonstration purposes. The containerized architecture translates directly to Kubernetes when scaling for production."

Q20: Why two PostgreSQL instances?

> "Best practice — Airflow's internal metadata (task states, DAG runs) shouldn't share a database with application data. If the warehouse has a heavy load, it shouldn't affect Airflow's ability to schedule tasks, and vice versa. Separating them prevents operational coupling."

Q21: How do you handle service dependencies?

> "Docker Compose health checks. PostgreSQL has pg_isready checks, MinIO has HTTP health endpoint checks. Airflow services have depends_on with condition: service_healthy, so they don't start until the databases are ready. This prevents startup race conditions."

Testing & Code Quality

Q22: How do you test a data pipeline?

> "I test at the unit level — each validation function, each transformation function, each detection algorithm. For example, I test that check_between correctly identifies values outside range, that standardize_columns renames correctly, that z-score detection catches obvious outliers. I have 38 tests covering ingestion config, quality validation, anomaly detection, and Silver transformations. Integration testing would be the next step — testing full pipeline runs with sample data."

Q23: Why pytest over unittest?

> "pytest is more Pythonic — less boilerplate, better assertions, better output. Test classes without inheriting from TestCase, simple assert statements instead of self.assertEqual, and excellent plugin ecosystem. It's the standard in the data engineering community."

Performance & Scale

Q24: How does your pipeline handle 9.5 million rows?

> "Batch inserts of 50,000 rows at a time (not one-by-one), SQL-based transformations for Gold layer (INSERT...SELECT processes all rows server-side without moving data through Python), streaming mode with vectorized pandas operations for validation. The Bronze and Silver loading are the

> bottleneck (~20-30 min each) because of Python-to-PostgreSQL data transfer. For larger scale, I'd use COPY command or Spark."

Q25: How would you scale this for 10x data?

> "Replace pandas with PySpark for distributed processing. Use COPY command instead of to_sql for PostgreSQL bulk loading. Move to a cloud warehouse like Snowflake that auto-scales. Add partitioning to PostgreSQL tables by date. Use Kafka partitions for parallel consumption. The architecture already supports these changes — it's a config/tool change, not a redesign."

General / Behavioral

Q26: What was the most challenging part?

> "Dependency management. Great Expectations conflicted with pandas/numpy versions, which led me to build a custom validation engine. SQLAlchemy versions conflicted with Airflow. This taught me that in production, pinning versions and managing dependencies is a real engineering challenge, not just a setup step."

Q27: What would you do differently?

> "I'd add dbt for SQL transformations — it provides version control, testing, and documentation for SQL models. I'd also add Spark for the heavy Silver/Bronze loading instead of pandas. And I'd implement incremental processing — right now the pipeline reprocesses all data; in production, you'd only process new/changed records."

Q28: What did you learn?

> "Three things: First, data quality is harder than it looks — 1.4% of real NYC data fails basic checks. Second, the architecture (Medallion, star schema) matters more than the specific tools — the patterns transfer across any stack. Third, containerization with Docker makes collaboration possible — anyone can run my entire pipeline with one command."

Q29: Why data engineering instead of data science?

> "Data science is exciting, but it depends on data engineering. The best ML model is useless if the data pipeline is unreliable, the data is dirty, or it can't scale. I enjoy building the infrastructure that makes analytics and ML possible — the plumbing that everything else depends on."

Q30: How does this project complement your other work?

> "FraudLens shows I can build ML models and analytics. SpendWiseAI shows I can build mobile applications. TaxiPulse shows I can build the data

> infrastructure that feeds both. Together, they demonstrate I can work across the full stack — from data pipelines to ML to user-facing apps."

Advanced / Deep Questions

Q31: Explain your idempotency strategy.

> "Every step checks if work was already done before proceeding. The download step skips already-downloaded files. The MinIO upload checks if objects exist. The Bronze loader checks source_file in the table. The Silver pipeline checks if the table has data. Gold builders check row counts. This means if the pipeline fails midway and restarts, it picks up where it left off without duplicating data."

Q32: How do you handle schema evolution?

> "Currently, the schema is fixed. For production, I'd add schema versioning — storing the expected schema in config and comparing against incoming data. If NYC TLC adds or renames a column, the validation step would catch it before bad data enters the pipeline. dbt also handles this well with its schema.yml testing."

Q33: What's the difference between ETL and ELT?

> "ETL transforms data before loading (Extract → Transform → Load). ELT loads raw data first, then transforms in-place (Extract → Load → Transform). My pipeline is more ELT — I load raw data into Bronze first, then transform within PostgreSQL. This is the modern pattern because warehouses are powerful enough to handle transformations, and keeping raw data gives you reprocessing flexibility."

Q34: How would you add data lineage?

> "I'd integrate OpenLineage — an open standard for tracking data lineage. Each Airflow task would emit lineage events showing input/output datasets. This would create a graph: 'NYC TLC file → Bronze table → Silver table → Gold fact_trips → agg_daily_summary → Streamlit dashboard.' Tools like Marquez or DataHub can visualize this lineage graph."

Q35: How do you ensure exactly-once processing in streaming?

> "Currently, the consumer uses auto-commit, which provides at-least-once delivery. For exactly-once, I'd use Kafka's transactional API — committing consumer offsets and database writes in the same transaction. Alternatively, I'd use a deduplication layer in Silver (checking event_id before insert). The current setup accepts that a few duplicates are acceptable for a demo."

Q36: What metrics would you monitor in production?

> "Pipeline metrics: task success rate, execution duration, data freshness (time since last successful load). Data quality metrics: pass rate trends, quarantine growth. Infrastructure metrics: PostgreSQL connection count, Kafka consumer lag, MinIO storage usage. Business metrics: daily trip volume, revenue totals (sudden drops indicate pipeline issues)."

Q37: How does your pipeline handle late-arriving data?

> "Currently, it doesn't — each run processes available data as-is. For production, I'd use event-time processing with watermarks. Late-arriving events in the streaming path would update existing aggregations. For batch, I'd implement a 'reconciliation window' — reprocessing the last N days to catch any late files."

Q38: What's the CAP theorem and how does it apply here?

> "CAP theorem states a distributed system can only guarantee two of three: Consistency, Availability, Partition tolerance. Kafka prioritizes availability and partition tolerance (AP) — consumers might read slightly stale data. PostgreSQL prioritizes consistency and partition tolerance (CP) — queries always return the latest committed data. My pipeline tolerates this because batch processing eventual consistency is fine, and the Gold layer is rebuilt periodically."

Q39: How would you implement data governance?

> "Column-level access control in PostgreSQL (GRANT/REVOKE on specific columns). Data classification labels (PII, sensitive, public) stored in a metadata catalog. Audit logging of who queries what data. Encryption at rest (PostgreSQL TDE) and in transit (SSL). For GDPR compliance, I'd add a data deletion pipeline that can purge specific records across all layers."

Q40: What's your data retention strategy?

> "Bronze: Keep forever (it's compressed Parquet, storage is cheap). Silver: Keep for the analysis window (1-2 years, then archive to cold storage). Gold aggregations: Keep forever (they're small). Quality and anomaly logs: Keep for compliance period. In the Docker setup, MinIO has a lifecycle rule that auto-deletes objects older than 90 days for cost control."

---

19. 30-Second Pitch

> "I built TaxiPulse — a Real-Time NYC Taxi Analytics Engine. It's an end-to-end data engineering platform that processes 9.5 million real taxi trip records through both batch and streaming pipelines. The batch path uses Airflow for orchestration, and the streaming path uses Kafka for real-time events. I implemented a Medallion Architecture with Bronze, Silver, and Gold layers, built

> a custom data quality engine with 18 automated checks achieving 98.6% pass rate, designed a star schema data warehouse, and created an anomaly detection system that found 3,623 fare and volume anomalies using Z-score analysis with Slack alerting. The entire infrastructure runs in Docker with 8 services, has 38 pytest tests with GitHub Actions CI/CD, and the dashboard is live at taxipulse-srujankothuri.streamlit.app."

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20. Key Numbers to Remember

What	Number
Total records	9,554,778
Quality pass rate	98.6%
Clean Silver records	9,417,374
Quarantined records	137,403
Quality checks	18
Total anomalies	3,623
Critical anomalies	1,478
Highest Z-score	20.6 (Zone 48, $81 avg fare at 4AM)
Star schema dimensions	5
Hourly aggregation rows	240,716
Daily summaries	93
Docker services	8
Airflow batch tasks	7
Pytest tests	38
NYC taxi zones	262
Data months	3 (Jan-Mar 2024)

---

21. "How I Built This" — Journey & Problems Solved

Timeline

Built in **~13 hours** across one intense session.

Step	What	Time	Key Challenge
Step 1	Project setup, folder structure,	30 min	Getting structure right from the start

Step	What	Time	Key Challenge
	config		
Step 2	Docker infrastructure (8 services)	2 hrs	Dependency conflicts — pandas vs great-expectations vs numpy vs SQLAlchemy vs Airflow. Took 3 rebuild cycles
Step 3	Data ingestion (download + MinIO upload)	45 min	SSL certificate errors on Mac for NYC TLC downloads
Step 4	Bronze layer loading (MinIO → PostgreSQL)	1 hr	Loading 9.5M rows in batches — tuning batch_size for memory
Step 5	Data quality validation engine	1 hr	Built custom engine after Great Expectations had dependency conflicts
Step 6	Silver layer transformation	30 min (code) + 30 min (waiting for 9.4M row insert)	Longest single operation — 30 min to write 9.4M rows
Step 7	Gold layer star schema	45 min	Foreign key violation — payment_type_id=0 not in dimension table
Step 8	Kafka streaming	45 min	Kafka port confusion — internal (9092) vs external (29092)
Step 9	Anomaly detection + Slack	30 min	Clean implementation, Z-score worked first try
Step 10	Streamlit dashboard	1.5 hrs	CSV fallback mode for cloud deployment was tricky
Step 11	Streamlit Cloud deployment	1 hr	CSV files not in git, TOML format for secrets, boolean parsing
Step 12	Bible Doc	1 hr	This document

Problems Faced & How I Solved Them

Problem 1: GCP Billing Verification Failed

What happened: Signed up for GCP $300 free trial, but the billing verification required a prepayment code from a credit card transaction. The code was truncated in the banking app and couldn't be entered.**Solution:** Pivoted to fully local Docker infrastructure with MinIO + PostgreSQL. This turned out to be a BETTER choice — zero cost, anyone can run it, demonstrates deeper Docker skills.**Lesson:** Always have a Plan B. Cloud dependency is a risk for portfolio projects.

Problem 2: Dependency Hell (pandas + great-expectations + numpy + Airflow)

What happened: Three consecutive Docker build failures:

1. great-expectations 1.3.0 requires pandas < 2.2, but we had pandas 2.2.3
2. After downgrading pandas to 2.1.4, numpy 2.1.3 conflicted (pandas 2.1.4 needs numpy < 2)
3. After fixing numpy, SQLAlchemy 2.0.36 conflicted with Airflow's internal SQLAlchemy version

Solution:

- Pinned compatible versions: pandas==2.1.4, numpy==1.26.4, pyarrow==14.0.2, scipy==1.12.0
- Removed SQLAlchemy pin from airflow requirements (let Airflow manage its own)
- Dropped great-expectations entirely and built a custom validation engine

Lesson: In production, dependency management is a real engineering problem. Pin versions, test compatibility, and don't be afraid to build custom solutions when libraries cause more problems than they solve.

Problem 3: Airflow Init Multiline Command Failure

What happened: The airflow users create command in docker-compose.yml broke because YAML multiline strings split the command arguments across lines, and bash interpreted each line as a separate command.**Solution:** Put the entire command on one line with && chaining.**Lesson:** YAML multiline strings and shell commands don't always play nicely. Keep critical commands on one line.

Problem 4: Foreign Key Violation in Gold Layer

What happened:fact_trips INSERT failed with: Key (payment_type_id)=(0) is not present in table "dim_payment_type". The Silver transformation filled null payment types with 0, but the dimension table only had IDs 1-6.**Solution:** Added payment_type_id=0 ("Unknown") and rate_code_id=0 ("Unknown") to the dimension tables.**Lesson:** When filling nulls with default values, make sure those defaults exist in referenced dimension tables. Always think about referential integrity.

Problem 5: Kafka Port Confusion

What happened: Kafka producer couldn't connect — KafkaTimeoutError: Failed to update metadata after 60.0 secs. The .env had KAFKA_BOOTSTRAP_SERVERS=localhost:9092.**Solution:** Changed to localhost:29092. Kafka exposes two listeners: 9092 for internal Docker network communication, 29092 for host machine access. Scripts running from the terminal

(outside Docker) need the external port.**Lesson:** Docker networking has internal vs external ports. Always check which port maps to what.

Problem 6: Streamlit CSV Fallback for Cloud Deployment

What happened: Streamlit Cloud can't access local PostgreSQL. Initial approach tried to parse SQL queries and route them to CSV files — this was brittle and broke on every page.**Solution:** Rewrote db.py with **named data functions** (get_bronze_count(), get_daily_summary(), etc.) that handle both PostgreSQL and CSV modes cleanly. Each function knows how to read from either source.**Lesson:** Don't try to be clever with SQL parsing. Use explicit, well-named functions that encapsulate the data access pattern.

Problem 7: Streamlit Auto-Detecting Pages

What happened: Streamlit automatically detected files in the pages/ folder and created its own navigation sidebar. This conflicted with the custom radio button navigation in app.py, causing Pipeline Overview to appear twice.**Solution:** Simplified app.py to just a landing page and let Streamlit's built-in page detection handle navigation.**Lesson:** Work WITH your framework, not against it. Streamlit's multi-page app feature is opinionated — embrace it.

---

22. Actual Performance Benchmarks

All benchmarks measured on **MacBook Air M2, 8GB RAM**, with Docker Desktop allocated 4GB RAM.

Pipeline Step Timings

Step	Duration	Rows Processed	Throughput
Download 3 Parquet files	~2 min	—	~75 MB/min (network dependent)
Upload to MinIO	~30 sec	3 files	~300 MB/min
Load Bronze → PostgreSQL	~12 min	9,554,778	~13,200 rows/sec
Data Quality Validation	~3 min	9,554,778	~53,000 rows/sec
Silver Transformation + Load	~25 min	9,417,374	~6,300 rows/sec (write bottleneck)
Gold Star Schema Build	~5 min	9,417,374 (fact)	SQL INSERT...SELECT
Gold Aggregations	~30 sec	240,716 + 93	SQL GROUP BY
Anomaly Detection	~2 sec	240,716 (hourly agg)	Near-instant

Step	Duration	Rows Processed	Throughput
Kafka Produce 500 events	~4 sec	500	~125 events/sec
Kafka Consume 500 events	~15 sec	500	~33 events/sec (includes DB writes)

Full Pipeline End-to-End

Metric	Value
Total pipeline time	~45-50 minutes
Bottleneck	Silver layer write (9.4M rows to PostgreSQL)
Fastest step	Anomaly detection (< 2 seconds)
Memory peak	~3-4 GB (loading 9.5M rows into pandas)

Dashboard Query Performance

Query	Source	Time
Pipeline Overview metrics	PostgreSQL (COUNT queries)	< 1 sec
Daily summary (93 rows)	PostgreSQL / CSV	< 100ms
Hourly aggregation (240K rows)	PostgreSQL	~500ms
Anomaly log (3,623 rows)	PostgreSQL / CSV	< 200ms
Full dashboard page load	All queries combined	1-3 sec

Where Would It Be Slow at Scale?

Scale	Bottleneck	Solution
100M rows	Pandas can't hold in memory	Use PySpark or Dask
100M rows	PostgreSQL INSERT too slow	Use COPY command or Spark JDBC
1B rows	Single PostgreSQL node	Move to Snowflake/Redshift (distributed)
10K events/sec streaming	Single Kafka consumer	Add consumer group with multiple consumers
1M anomaly checks	Z-score on full dataset	Pre-compute rolling statistics, incremental checks

---

23. Code Walkthrough — Key Snippets Explained

1. Data Quality — Running an Expectation

```
def check_between(df, column, min_val, max_val):
    """Returns True for rows where column is within [min, max]."""
    col = df[column]
    return (col.isna()) | ((col >= min_val) & (col <= max_val))
```

Why it works this way:

- col.isna() — Null values PASS range checks (null checks are separate). This prevents double-penalizing nulls.
- Uses **vectorized pandas operations** — no Python for-loop. Processes 9.5M rows in milliseconds.
- Returns a boolean Series (mask) — True = passes, False = fails. Masks are combined later to determine which rows to quarantine.

2. Silver Transformation — Adding Derived Columns

```
def add_derived_columns(df):
    df["trip_duration_minutes"] = (
        (df["dropoff_datetime"] - df["pickup_datetime"])
        .dt.total_seconds() / 60
    ).round(2)
    df["trip_duration_minutes"] = df["trip_duration_minutes"].clip(lower=0,
        upper=1440)
    return df
```

Why clip?

- Some trips have dropoff BEFORE pickup (data quality issue we didn't quarantine if it's a warning)
- Some trips span multiple days (legitimate but extreme)
- clip(lower=0, upper=1440) caps at 0–24 hours — reasonable for any taxi trip
- Without clipping, negative durations or multi-day durations would skew averages

3. Gold Layer — SQL INSERT...SELECT for Fact Table

```
conn.execute(text("""
    INSERT INTO gold.fact_trips (
        datetime_id, pickup_location_id, dropoff_location_id,
        payment_type_id, rate_code_id,
        passenger_count, trip_distance, trip_duration_minutes,
```

```
        fare_amount, tip_amount, tolls_amount,
        total_amount, congestion_surcharge, airport_fee
    )
    SELECT
        d.datetime_id,
        s.pickup_location_id,
        s.dropoff_location_id,
        ...
    FROM silver.clean_yellow_trips s
    JOIN gold.dim_datetime d
        ON date_trunc('hour', s.pickup_datetime) = d.full_datetime
"""))
```

Why SQL instead of Python?

- This single SQL statement processes ALL 9.4M rows server-side within PostgreSQL
- No data moves through Python — no memory issues, no network overhead
- The JOIN with dim_datetime maps each trip's pickup hour to a datetime_id
- date_trunc('hour', ...) truncates to the hour — matching the granularity of our dimension
- This is 10-100x faster than reading into pandas, joining in Python, and writing back

4. Anomaly Detection — Z-Score Calculation

```python
def detect_zscore_anomalies(df, value_col, group_col=None, threshold=3.0):
    if group_col and group_col in df.columns:
        grouped = df.groupby(group_col)[value_col]
        df["mean"] = grouped.transform("mean")
        df["std"] = grouped.transform("std")
    else:
        df["mean"] = df[value_col].mean()
        df["std"] = df[value_col].std()

    df["std"] = df["std"].replace(0, np.nan)  # Avoid division by zero
    df["z_score"] = ((df[value_col] - df["mean"]) / df["std"]).abs()
    anomalies = df[df["z_score"] >= threshold].copy()
    return anomalies
```

Key design choices:

- groupby().transform() — Calculates mean/std PER GROUP but broadcasts back to original DataFrame size. Zone 48's mean is different from Zone 132's mean.
- replace(0, np.nan) — If a zone has zero variance (all same values), std=0 would cause division by zero. NaN prevents false positives.
- .abs() — Catches anomalies in BOTH directions (unusually high AND unusually low)
- Returns a copy — doesn't modify the original DataFrame

5. Kafka Producer — Event Publishing

```
event = {
    **trip,                                          # All trip fields
    "event_id": f"evt-{int(time.time()*1000)}-{i}", # Unique ID
    "event_timestamp": datetime.now().isoformat(),   # When event was created
    "event_type": "trip_completed",                  # Event classification
}
key = str(trip.get("pickup_location_id", "unknown"))
producer.send(topic, value=event, key=key)
```

Why partition by pickup_location_id?

- Kafka partitions messages by key — same key always goes to same partition
- All events from Zone 132 go to the same partition → preserved ordering within a zone
- Consumers can process zone-specific data without cross-partition coordination
- In production, this enables parallel consumers per zone group

6. Idempotent Loading — Skip Already-Processed Data

```
def check_already_loaded(engine, source_file):
    query = text(
        "SELECT COUNT(*) FROM bronze.raw_yellow_trips "
        "WHERE source_file = :file"
    )
    with engine.connect() as conn:
        result = conn.execute(query, {"file": source_file}).scalar()
    return result > 0
```

Why this matters:

- If the pipeline crashes at Step 5 and restarts, it shouldn't re-load Steps 1-4
- Each file has a source_file column tracking its origin
- Before loading, we check if that file is already in the table
- Same pattern in every step: download (check file exists), upload (check MinIO object exists), Silver (check table has data), Gold (check row count)
- This is **idempotency** — running the pipeline multiple times produces the same result

---

24. Comparison — Why TaxiPulse Stands Out

vs. Typical Tutorial Projects

Feature	Typical Tutorial	TaxiPulse
Data volume	10K-100K rows	**9.5M rows**
Data source	CSV file	**Real NYC TLC Parquet files**
Ingestion	Single batch path	**Batch + Kafka streaming**
Data quality	None	**18 automated checks + quarantine**
Architecture	Flat table	**Medallion (Bronze → Silver → Gold)**
Data modeling	None	**Star schema (fact + 5 dimensions)**
Anomaly detection	None	**Z-score on 3 dimensions + Slack alerts**
Orchestration	Manual scripts	**Airflow DAGs (7+2 tasks)**
Testing	None	**38 pytest tests**
CI/CD	None	**GitHub Actions**
Deployment	Local only	**Live Streamlit Cloud dashboard**
Documentation	README	**Complete Bible Doc**

vs. Darshil Parmar's Uber ETL (Most Popular DE Project)

Feature	Uber ETL Project	TaxiPulse
Pipeline type	Batch only	**Batch + Streaming**
Data quality	None	**18 checks + quarantine**
Anomaly detection	None	**Z-score + Slack alerting**
Data warehouse	BigQuery (cloud)	PostgreSQL (local, cloud-agnostic)
Orchestration	Mage	**Airflow (industry standard)**
Streaming	None	**Kafka (producer + consumer)**
Testing	None	**38 tests + CI/CD**
Runnable by anyone	Needs GCP account	**Just needs Docker**
Dashboard	Looker Studio	**Streamlit (custom, deployed)**

vs. airscholar/e2e-data-engineering (Popular GitHub Project)

Feature	airscholar Project	TaxiPulse
Data source	RandomUser API (fake)	**Real NYC TLC data (9.5M rows)**
Data quality	None	**18 checks + quarantine**
Anomaly detection	None	**Z-score + alerting**

Feature	airscholar Project	TaxiPulse
Star schema	None	**Fact + 5 dimensions + aggregations**
Testing	None	**38 tests + CI/CD**
Live demo	None	**Streamlit Cloud URL**

The "Extra 20%" That Makes It Novel

Most projects have: ETL pipeline, cloud storage, warehouse, dashboard. TaxiPulse adds: **streaming + quality engine + anomaly detection + alerting + tests + CI/CD + deployed dashboard.** This combination is rare in portfolio projects and directly maps to what companies build in production.

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25. Scenario-Based Interview Questions

These are "What if..." questions that test production thinking.

Q: What if the NYC TLC website goes down during a batch run?

> "The download task would fail after the request timeout. Airflow retries the task twice (configured retry_delay=5min). If it still fails, the task is marked as failed and downstream tasks don't run — Bronze/Silver/Gold from previous runs are unaffected. I'd also add a monitoring alert for consecutive download failures."

Q: What if someone uploads a corrupted Parquet file?

> "PyArrow would throw an error when reading the file, which the download step catches. The file wouldn't be loaded to Bronze. If somehow a corrupted file gets through to Bronze, the data quality validation in Step 4 would catch anomalous values — invalid dates, out-of-range fares, null critical fields. Those records would be quarantined."

Q: What if the data schema changes (NYC TLC adds a new column)?

> "The Bronze loader uses standardize_columns() which only keeps columns in a predefined mapping. A new column would be silently ignored — not ideal. For production, I'd add schema validation as a quality check: compare incoming columns against expected columns and alert if there's a mismatch. This gives the team time to update the pipeline."

Q: What if Kafka goes down while the producer is sending events?

> "The producer has retries=3 configured, so transient failures are retried. If Kafka is completely down, the producer raises an exception after retries. In production, I'd add a dead letter queue — failed events go to a backup store

> (S3/MinIO) and are replayed when Kafka recovers. The batch path provides a safety net for any streaming data loss."

Q: What if a zone suddenly has 100x normal fare (real surge pricing)?

> "The anomaly detection would flag it as a fare_spike with a high z-score. The question is: is it a data error or legitimate surge? I'd add context to the anomaly — check if volume also spiked (event like a concert or holiday). If high fare + high volume → legitimate surge. If high fare + normal volume → possible data error. The system detects it; human judgment interprets it."

Q: What if the PostgreSQL database runs out of disk space?

> "Docker volumes have no built-in size limits, so it would eventually fill the host disk. For production: monitor disk usage with alerts at 70%/85%/95%, implement data retention policies (archive Bronze data older than 6 months to cold storage), partition large tables by month and drop old partitions, use PostgreSQL's VACUUM to reclaim space from deleted rows."

Q: What if two Airflow runs overlap (previous run still executing when new one triggers)?

> "The DAG has depends_on_past=False and catchup=False, so Airflow won't auto-trigger overlapping runs. But if manually triggered: the idempotency checks prevent duplication — each step checks if its work is already done. The bigger risk is resource contention (two runs fighting for memory). In production, I'd use Airflow's max_active_runs=1 to prevent this."

Q: What if a critical anomaly is detected at 3 AM?

> "Currently, anomaly detection runs as the last Airflow task, so it depends on the batch schedule. The Slack alert would fire immediately when the task runs. For true 24/7 alerting, I'd run the anomaly detection on a separate, more frequent schedule (every hour) against the Gold aggregations. Critical severity anomalies could trigger PagerDuty instead of just Slack."

Q: What if you need to reprocess all data from scratch?

> "That's exactly why Bronze exists. Steps: 1) Truncate Silver and Gold tables, 2) Re-run the pipeline from the validate step onward. Bronze is untouched — it's the raw source of truth. The pipeline's idempotency means I'd need to clear the 'already processed' markers (truncate Silver, Gold), then re-run. Total reprocess time: ~45 minutes for 9.5M rows."

Q: What if a new data engineer joins and needs to understand the pipeline?

> "They'd read this Bible Doc, then run make pipeline to see the full flow. The Docker Compose setup means they don't need to install anything except Docker. Airflow UI shows the DAG visually with task dependencies. The

> Streamlit dashboard shows the output. The tests serve as documentation of expected behavior. And the code has docstrings and logging throughout."

Q: What if you need to add a new data source (e.g., weather data)?

> "I'd add a new ingestion module under ingestion/batch/, a new Bronze table, new quality expectations, and a new Silver table. The Gold layer would get a new dimension (dim_weather) and the fact table would gain a weather_id foreign key. The Airflow DAG would get new tasks for the weather path, running in parallel with the taxi data tasks. The Medallion architecture makes this clean — new sources get their own Bronze/Silver path but merge at Gold."

Q: What if the anomaly threshold is too sensitive (too many false positives)?

> "Tune the Z-score threshold. Currently 3.0 detects 3,623 anomalies. Raising to 4.0 would reduce to ~1,478 (only critical). I'd analyze the false positive rate by sampling flagged anomalies and checking if they're genuine. Another approach: use IQR (Interquartile Range) method alongside Z-score and only flag anomalies detected by BOTH methods, reducing false positives."

---

26. Troubleshooting Guide

Docker Issues

Problem: docker-compose up fails with port already in use

```
Error: Bind for 0.0.0.0:5432 failed: port is already allocated
```

Fix: Another PostgreSQL is running locally. Either stop it (brew services stop postgresql) or change the port in docker-compose.yml (e.g., 5434:5432).

Problem: Containers keep restarting

```
STATUS: Restarting (1) 23 seconds ago
```

Fix: Check logs: docker-compose logs <service-name>. Usually a dependency or configuration issue. Common fix: docker-compose down -v then docker-compose up -d for a clean start.

Problem: Airflow webserver won't startFix: Usually the init container didn't run properly. Check: docker-compose logs airflow-init. If it failed, run docker-compose down -v and restart.

Problem: Out of disk spaceFix:docker system prune -a removes unused images/containers. docker volume prune removes unused volumes. Check Docker Desktop settings for disk usage.

Python Issues

Problem: ModuleNotFoundError: No module named 'X'Fix: Make sure your venv is activated (source venv/bin/activate) and install the missing package (pip install X).

Problem: Streamlit uses system Python instead of venvFix: Use python -m streamlit run instead of just streamlit run. This ensures it uses the venv's Python.

Problem: SSL certificate error on MacFix: Use requests library with verify=False instead of pd.read_csv(url) for HTTPS downloads.

Database Issues

Problem: psycopg2.errors.ForeignKeyViolationFix: A value in the fact table doesn't exist in the dimension table. Add the missing value to the dimension table (e.g., INSERT INTO gold.dim_payment_type (payment_type_id, payment_description) VALUES (0, 'Unknown')).

Problem: Data seems duplicated after re-running pipelineFix: The pipeline has idempotency checks, but if you manually inserted data, truncate and re-run: TRUNCATE silver.clean_yellow_trips CASCADE; then python scripts/run_silver.py.

Kafka Issues

Problem: KafkaTimeoutError: Failed to update metadataFix: You're using the wrong port. From terminal (outside Docker), use localhost:29092. Inside Docker containers, use kafka:9092.

Problem: Consumer gets no messagesFix: Check the topic exists: docker exec taxipulse-kafka kafka-topics --list --bootstrap-server localhost:9092. If empty, produce some events first.

Streamlit Cloud Issues

Problem: Dashboard shows no data after deploymentFix: CSV files not committed to git. Run: git add -f streamlit_app/data/*.csv && git commit -m "Add CSV data" && git push

Problem: KeyError on Streamlit Cloud but works locallyFix: CSV column names may differ from SQL column names. Check the CSV headers match what the code expects. The db.py named functions handle this — make sure you're using the latest version.

---

27. Project File Structure — Every File Explained

```
TaxiPulse/
│
├── docker-compose.yml              # Defines all 8 Docker services
(PostgreSQL, MinIO, Kafka, Airflow, etc.)
├── docker/
│   └── Dockerfile.airflow          # Custom Airflow image with extra Python
packages
│
├── requirements.txt                # Python dependencies for local development
├── requirements-airflow.txt        # Python dependencies for Airflow Docker
container
├── pytest.ini                      # Pytest configuration (test paths,
verbosity)
├── Makefile                        # Shortcut commands (make pipeline, make
test, etc.)
├── .env.example                    # Template for environment variables (copy
to .env)
├── .env                            # YOUR actual environment variables (never
committed)
├── .gitignore                      # Files excluded from Git
│
├── config/                         # Centralized configuration
│   ├── __init__.py
│   ├── settings.py                 # Loads .env variables into typed Python
classes
│   │                               #   MinIOConfig, PostgresConfig,
KafkaConfig, etc.
│   └── constants.py                # Static values — column mappings, payment
types,
│                                   #   rate codes, quality thresholds, table
names
│
├── ingestion/                      # Data ingestion (getting data INTO the
pipeline)
│   ├── __init__.py
│   ├── batch/                      # Batch ingestion path
│   │   ├── __init__.py
│   │   ├── download_tlc_data.py    # Downloads Parquet files from NYC TLC
website
│   │   └── upload_to_minio.py      # Uploads Parquet files to MinIO bronze/
bucket
│   └── streaming/                  # Streaming ingestion path
│       ├── __init__.py
│       ├── kafka_producer.py       # Reads from Silver, publishes events to
Kafka topic
│       └── kafka_consumer.py       # Consumes Kafka events, validates, inserts
to Silver
│
├── transformations/                # Data transformation (Bronze → Silver →
Gold)
│   ├── __init__.py
```

```
│   ├── bronze/
│   │   ├── __init__.py
│   │   └── load_raw_to_postgres.py # Reads Parquet from MinIO, inserts to
bronze.raw_yellow_trips
│   ├── silver/
│   │   ├── __init__.py
│   │   └── clean_and_validate.py   # Validates Bronze data, transforms clean
rows to Silver,
│   │                               #   quarantines failed rows. Adds
trip_duration_minutes,
│   │                               #   standardizes column names, enforces
types, deduplicates
│   └── gold/
│       ├── __init__.py
│       └── build_star_schema.py    # Builds dim tables, fact_trips,
agg_hourly_zone_revenue,
│                                   #   agg_daily_summary. All via SQL
INSERT...SELECT
│
├── quality/                        # Data quality validation engine
│   ├── __init__.py
│   ├── validate_data.py            # Core engine — runs expectations, splits
clean/quarantine,
│   │                               #   logs results to gold.quality_log
│   └── expectations/
│       ├── __init__.py
│       └── taxi_expectations.py    # Defines all 18 expectations (null, range,
categorical,
│                                   #   temporal) with severity levels and
descriptions
│
├── anomaly_detection/              # Anomaly detection + alerting
│   ├── __init__.py
│   ├── detector.py                 # Z-score analysis on Gold aggregations.
Detects fare spikes,
│   │                               #   volume spikes, daily revenue anomalies.
Logs to anomaly_log
│   └── alerting.py                 # Sends Slack alerts via webhook. Falls
back to console if
│                                   #   Slack not configured. Marks alerts as
sent in DB
│
├── airflow/                        # Apache Airflow configuration
│   ├── dags/
│   │   ├── taxipulse_batch_dag.py  # 7-task batch DAG: download → upload →
bronze → validate
│   │   │                           #   → silver → gold → anomaly detection
│   │   └── taxipulse_streaming_dag.py  # 2-task streaming demo DAG: produce →
consume
│   ├── logs/                       # Airflow task execution logs (gitignored)
│   ├── plugins/                    # Custom Airflow plugins (empty — for
extensions)
│   └── config/                     # Airflow config overrides
│
├── streamlit_app/                  # Monitoring dashboard
│   ├── app.py                      # Main entry point — landing page with
project highlights
```

```
│   ├── db.py                       # Data access layer — named functions that
work in both
│   │                               #   PostgreSQL mode (local) and CSV mode
(Streamlit Cloud)
│   ├── requirements.txt            # Streamlit Cloud dependencies (subset of
main requirements)
│   ├── .streamlit/
│   │   └── config.toml             # Streamlit theme (dark mode, colors)
│   ├── data/                       # Exported CSV files for Streamlit Cloud
deployment
│   │   ├── agg_daily_summary.csv
│   │   ├── agg_hourly_zone_revenue.csv
│   │   ├── anomaly_log.csv
│   │   ├── quality_log.csv
│   │   ├── dim_pickup_location.csv
│   │   ├── bronze_count.csv
│   │   ├── silver_count.csv
│   │   ├── quarantine_count.csv
│   │   ├── fact_count.csv
│   │   ├── dim_datetime_count.csv
│   │   └── freshness.csv
│   └── pages/                      # Dashboard pages (auto-detected by
Streamlit)
│       ├── __init__.py
│       ├── pipeline_overview.py    # Key metrics, pass rate, Gold stats,
architecture diagram
│       ├── analytics_explorer.py   # Revenue trends, hourly patterns, top
zones, payments
│       ├── anomaly_monitor.py      # Anomaly counts, filters, zone chart, z-
score distribution
│       └── data_quality_report.py  # Pass rates, per-check breakdown,
quarantine reasons
│
├── scripts/                        # Standalone runner scripts
│   ├── init_db.sql                 # PostgreSQL init — creates schemas, all
tables, inserts
│   │                               #   dimension seed data. Runs automatically
on first container start
│   ├── run_ingestion.py            # Runs: download + upload + bronze load
│   ├── run_quality_check.py        # Runs: data quality validation only
│   ├── run_silver.py               # Runs: silver transformation pipeline
│   ├── run_gold.py                 # Runs: gold layer build
│   ├── run_anomaly_detection.py    # Runs: anomaly detection + alerting
│   ├── run_streaming_demo.py       # Runs: kafka produce + consume demo
│   ├── load_zone_names.py          # Downloads NYC zone lookup, updates
dimension tables
│   └── run_pipeline.sh             # Shell script placeholder
│
├── tests/                          # Pytest test suite (38 tests)
│   ├── __init__.py
│   ├── test_ingestion/
│   │   ├── __init__.py
│   │   └── test_download.py        # 5 tests — TLC config, URL format,
filename generation
│   ├── test_quality/
│   │   ├── __init__.py
│   │   └── test_validation.py      # 13 tests —
```

```
null/range/categorical/temporal checks, full validation
│   ├── test_anomaly/
│   │   ├── __init__.py
│   │   └── test_detector.py        # 6 tests — z-score detection, grouping,
thresholds, edge cases
│   └── test_transformations/
│       ├── __init__.py
│       └── test_silver.py          # 8 tests — column rename, type
enforcement, duration, dedup (was 14 - reduced)
│
├── terraform/                      # Infrastructure as Code (placeholder —
project uses Docker)
│   ├── main.tf
│   ├── variables.tf
│   ├── outputs.tf
│   └── provider.tf
│
├── docs/                           # Documentation
│   ├── architecture.md
│   └── setup_guide.md
│
├── .github/
│   └── workflows/
│       └── test.yml                # GitHub Actions CI — runs tests + syntax
check on every push
│
└── README.md                       # Project overview, architecture diagram,
tech stack, quick start
```

---

28. Environment Variables Reference

Every variable in .env explained:

MinIO (Object Storage)

Variable	Default	Description
MINIO_ENDPOINT	minio:9000 (Docker) / localhost:9000 (local)	MinIO API endpoint. Use minio:9000 inside Docker, localhost:9000 from terminal
MINIO_ACCESS_KEY	taxipulse	MinIO username (like AWS Access Key)
MINIO_SECRET_KEY	taxipulse123	MinIO password (like AWS Secret Key)
MINIO_BUCKET_NAME	taxipulse-data-lake	Name of the data lake bucket
MINIO_BRONZE_PREFIX	bronze/	Folder prefix for raw data in bucket
MINIO_SILVER_PREFIX	silver/	Folder prefix for cleaned data
MINIO_GOLD_PREFIX	gold/	Folder prefix for business-ready data

Variable	Default	Description
MINIO_USE_SSL	false	Whether to use HTTPS (false for local Docker)

PostgreSQL (Data Warehouse)

Variable	Default	Description
POSTGRES_HOST	postgres (Docker) / localhost (local)	Database hostname
POSTGRES_PORT	5432	Database port
POSTGRES_DB	taxipulse	Database name
POSTGRES_USER	taxipulse	Database username
POSTGRES_PASSWORD	taxipulse123	Database password
PG_SCHEMA_BRONZE	bronze	Schema for raw data
PG_SCHEMA_SILVER	silver	Schema for cleaned data
PG_SCHEMA_GOLD	gold	Schema for business-ready data

NYC TLC Data Source

Variable	Default	Description
TLC_BASE_URL	https://d37ci6vzurychx.cloudfront.net/trip-data	Base URL for NYC TLC Parquet files
TLC_DATA_YEAR	2024	Year of data to download
TLC_DATA_MONTHS	01,02,03	Comma-separated months to download (add more for more data)

Apache Kafka

Variable	Default	Description
KAFKA_BOOTSTRAP_SERVERS	kafka:9092 (Docker) / localhost:29092 (local)	Kafka broker address. **IMPORTANT:** Use port 29092 from terminal, 9092 inside Docker
KAFKA_TOPIC	taxipulse-trips	Topic name for taxi trip events

Slack Alerting

Variable	Default	Description
SLACK_WEBHOOK_URL	(placeholder)	Slack incoming webhook URL. Get from api.slack.com → Your App → Incoming Webhooks
SLACK_CHANNEL	#taxipulse-alerts	Slack channel for anomaly alerts

Anomaly Detection

Variable	Default	Description
ANOMALY_ZSCORE_THRESHOLD	3.0	Z-score threshold for flagging anomalies. Lower = more sensitive, higher = fewer alerts
ANOMALY_CHECK_INTERVAL_HOURS	1	How often to check (used in production scheduling)

Streamlit

Variable	Default	Description
STREAMLIT_PORT	8501	Port for local Streamlit dashboard
USE_CSV	false	Set to true on Streamlit Cloud (reads from CSV instead of PostgreSQL)

Key Rule: Docker Internal vs External

Service	Inside Docker (docker-compose)	Outside Docker (terminal)
MinIO	minio:9000	localhost:9000
PostgreSQL	postgres:5432	localhost:5432
Kafka	kafka:9092	localhost:29092

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29. Skills Demonstrated — Component to Skill Map

When a recruiter reads your resume or a hiring manager reviews your project, here's exactly what each component signals:

Technical Skills

Component You Built	Skill It Demonstrates	Resume Keyword
NYC TLC Parquet ingestion	Working with real-world data at scale	Data Ingestion, ETL
MinIO data lake (bronze/)	Object storage / Data Lake design	S3, Data Lake, Object Storage
PostgreSQL warehouse	Relational database management	SQL, Data Warehousing, PostgreSQL
Bronze → Silver → Gold	Medallion Architecture	Data Architecture, ELT
Star schema (fact + dims)	Dimensional modeling	Data Modeling, Star Schema, Kimball
Pre-computed aggregations	Query optimization	Performance Optimization
18-check quality engine	Data quality engineering	Data Quality, Data Validation
Quarantine table	Error handling & data governance	Data Governance
Z-score anomaly detection	Statistical analysis	Anomaly Detection, Statistics
Slack webhook alerting	Production monitoring	Alerting, Observability
Kafka producer/consumer	Event-driven architecture	Apache Kafka, Streaming, Event-Driven
Airflow DAGs (7+2 tasks)	Pipeline orchestration	Apache Airflow, Orchestration, DAGs
Docker Compose (8 services)	Containerization & infrastructure	Docker, Containerization, IaC
Streamlit dashboard (4 pages)	Data visualization	Dashboarding, Streamlit, Plotly
38 pytest tests	Testing & code quality	pytest, Unit Testing, TDD
GitHub Actions CI/CD	Continuous integration	CI/CD, GitHub Actions, DevOps
Makefile commands	Developer experience	Automation, DevOps
Config management (.env)	Secure configuration	Configuration Management
Idempotent pipeline design	Production reliability	Reliability Engineering

Soft Skills

What You Did	Skill It Shows
Pivoted from GCP to local Docker when billing failed	**Adaptability, Problem-solving**

What You Did	Skill It Shows
Built custom quality engine when Great Expectations had conflicts	**Independent problem-solving, Engineering judgment**
Processed 9.5M real rows (not toy data)	**Working at scale**
Comprehensive documentation (Bible Doc, README)	**Technical communication**
12-step pipeline with proper error handling	**Systems thinking**
Designed for both batch and streaming	**Architectural thinking**

Role Fit Matrix

Role	Relevant Components	Fit Score
Data Engineer	Airflow, Kafka, Medallion, Star Schema, Docker, Quality Engine	⭐⭐⭐⭐⭐
Data Platform Engineer	Docker, Airflow, Kafka, PostgreSQL, MinIO, CI/CD	⭐⭐⭐⭐⭐
Analytics Engineer	Star Schema, Aggregations, Streamlit, SQL	⭐⭐⭐⭐
ML Engineer	Data pipeline, Quality validation, Feature engineering (Silver)	⭐⭐⭐⭐
Backend Engineer	Docker, PostgreSQL, Kafka, Python, Testing, CI/CD	⭐⭐⭐⭐
Full Stack Developer	Streamlit, PostgreSQL, Docker, Python, Git	⭐⭐⭐
Data Scientist	Anomaly detection, Statistical analysis, Data quality	⭐⭐⭐
DevOps/SRE	Docker, CI/CD, Monitoring, Alerting, Infrastructure	⭐⭐⭐

Combined Portfolio Story

Project	Role Signal	Domain
FraudLens	Data Science, ML, Analytics	Finance/Security
SpendWiseAI	Mobile Development, Full-Stack	Consumer/FinTech
TaxiPulse	Data Engineering, Infrastructure	Transportation/Analytics

Together: "I can build ML models (FraudLens), ship user-facing apps (SpendWiseAI), AND build the data infrastructure that powers both (TaxiPulse). I work across the full stack."

This document is your complete reference for understanding, explaining, and defending the TaxiPulse project in any interview setting. Study the Q&A section, know your numbers, rehearse the "How I Built This" stories, and you'll confidently handle anything they throw at you.

Total: 29 sections | 50+ interview Q&As | Performance benchmarks | Code walkthroughs | Scenario questions | Troubleshooting guide